

Received 17 July 2024, accepted 7 August 2024, date of publication 9 August 2024, date of current version 20 August 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3441094

RESEARCH ARTICLE

Securing Consumer Electronics Devices: A Blockchain-Based Access Management Approach Enhanced by Deep Learning Threat Modeling for IoT Ecosystems

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The authors extend their appreciation to the Deanship of Research and Graduate Studies at King Khalid University for funding this work through Large Research Project under grant number RGP2/43/45. Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2024R411), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia. The authors extend their appreciation to the Deanship of Scientific Research at Northern Border University, Arar, KSA for funding this research work through the project number NBU-FFR-2024-3030-03. This study is partially funded by the Future University in Egypt (FUE).

ABSTRACT Securing user electronics devices has become a significant concern in the digital period, and a forward-thinking solution covers the fusion of blockchain (BC) technology and deep learning (DL) methods. Blockchain improves device safety by transforming access management, storing credentials on a tamper-resistant ledger, mitigating the risk of unauthorized access and giving a robust defence against malevolent actors. Integrating DL into this framework also raises safety measures, as it permits devices to inspect and regulate to develop attacks distinctly. DL models accurately recognize intricate designs and anomalies, allowing the technique to distinguish and threaten possible attacks in real time. The fusion of BC and DL not only improves the reliability of user electronics but also establishes a dynamic and adaptive safety system, enhancing consumer confidence in the safety of their devices. Therefore, this study presents a BC-Based Access Management with DL Threat Modeling (BCAM-DLTM) technique for securing consumer electronics devices in the IoT ecosystems. The BCAM-DLTM technique mainly follows a two-phase procedure: access management and threat detection. Moreover, BC technology can be applied to the access management of consumer electronics devices. Besides, the BCAM-DLTM technique applies a deep belief networks (DBNs) model for proficiently identifying threats. To enhance the recognition results of the DBN model, the hyperparameter tuning procedure uses the reptile search algorithm (RSA). The experimental outcome study of the BCAM-DLTM approach employs the NSLKDD dataset. The comprehensive results of the BCAM-DLTM approach portrayed a superior accuracy outcome of 99.63% over existing models in terms of distinct metrics.

INDEX TERMS Blockchain, Internet of Things, deep learning, reptile search algorithm, deep belief network.

I. INTRODUCTION

The growth of industrial applications of IoT and networking facilities has simplified a significant rise in the integer

The associate editor coordinating the review of this manuscript and approving it for publication was Sedat Akleylek¹.

of linked devices [1]. These application devices can take accurate trade data with a keen sensor unit. Industrial development and technical control are behind this change in how systems interrelate with logical and physical things [2]. A central architecture is utilized to connect accurate industrial data and assess the crucial components of IoT with

self-management. The market of user electronics is influenced by the technology of IoT [3]. More home appliances are prepared by communication interfaces like Bluetooth, Ethernet, or WLAN, which permit uniting and handling within a local area network (LAN). Besides, many of them are prepared by an innovative operating method, enabling dynamic management, e.g., by setting the processing time by considering energy tariffs or restrained environmental states like humidity or temperature [4].

Data security is a primary concern of IoT systems because they are made by linking numerous IoT devices [5]. Data produced by these devices are kept in the cloud and spread across several networks. A cyberattack on an intelligent healthcare method can significantly control the system's capability to generate and supply energy. Also, to economic and other harm, cyberattacks can cause practical failures, theft of crucial data, power outages, and whole safety breaks. Cyber specialists face problems with protection labels on all licenses through an intelligent network and identifying possible attacks and threats [6]. While machine learning (ML) has become a crucial part of cybersecurity, the issue is that this range needs separate methods and theoretical views to manage the enormous data size produced and conveyed through many networks in an intelligent network. The threats and attacks that could be initiated beside this proof of model environment are being determined utilizing threat modelling [7]. Numerous possible threats have been verified with recognition, repudiation, tampering, data leakage, extended privilege (EoP), and denial of service (DoS). Each hazard and the safety elements linked with them are referred to utilizing STRIDE. STRIDE is a usual threat-modelling model for discovering and categorizing attack vectors. Researchers can use MITRE ATTACK's recognized trade framework to identify threats masked as tactics, techniques, and procedures (TTP).

In the present scenario, it has been detected that BC technology and consensus protocols play an essential part in safeguarding IoT environments owing to their absolute and tamper-resistant information [8]. BC delivers a distributed and decentralized ledger that can stock IoT data immutable and transparently [9]. By keeping data in blocks and connecting them with cryptographic hashes, the immutability and reliability of the data can be certified [10]. This helps in preventing data varying and illegal changes. Consensus protocols, namely Proof of Stake (PoS) or Proof of Work (PoW), defend contracts and trust in a BC network [11]. These protocols permit nodes in the method to spread a consensus on the authority of contacts and the state of the BC [12]. By employing consensus devices, IoT environments can verify data consistency and avoid malicious actions [13]. This study utilizes CNNs for embedding encrypted messages into multi-media [14], which BC enhances for immutable data recording, addressing resource restrictions with lightweight techniques in DeepStegBlock [15].

This study proposes a BC-Based Access Management with DL Threat Modeling (BCAM-DLTM) method for safeguarding user electronics devices in the IoT ecosystems. The BCAM-DLTM method primarily follows a dual-phase method, which includes access management and threat recognition. In the BCAM-DLTM approach, the BC model can manage access to user electronic devices. Besides, the BCAM-DLTM technique applies a deep belief networks (DBNs) model for proficiently identifying threats. To enhance the recognition results of the DBN technique, the hyperparameter tuning progress takes place utilizing the reptile search algorithm (RSA). The experimental solution study of the BCAM-DLTM technique takes place using a benchmark database.

- This study presents a novel BCAM-DLTM model that transforms consumer electronics management by utilizing BC to improve access control and safety. It provides a transparent and immutable credential management system to enhance device safety.
- The presented technique detects and addresses consumer electronics threats by integrating DBNs and BC, utilizing pattern detection of DBNs to protect against safety breaches and malevolent activities proactively, thus confirming the integrity and safety of the model
- The model also presents the RSA method for fine-tuning the hyperparameters, which is significant for optimizing ML accomplishment. This technique automatically adjusts model parameters, thus enhancing the accuracy and efficiency of threat recognition and management
- This approach, integrating BC, DBNs, and the RSA, marks a significant leap in consumer electronics management and safety. It gives a holistic outcome to modern device threats, improving consumers' safety and dependability.

II. LITERATURE SURVEY

Guduri et al. [16] presented a BC-based lightweight encryption scheme with federated learning (FL). The model builds up dynamic smart contracts in run-time among the sensor as well as data consumer to enable the transmission of EHR data. The technique utilizes a highly efficient proxy re-encryption method with FL. In [17], DL, BC, and fully homomorphic encryption (FHE) techniques in VANETs developed a Decentralized Privacy-preserving DL (DPDL) method that has been implemented. The technique also presented a Decentralized VANETs (DVANETs) method. The method employs BC and introduces a DPDL method for offering privacy-preservation data analysis for DVANET, wherein the FHE technique was used to encrypt the transmitted data. In [18], a learning technique dependent upon enabled DL and BC methods was presented and developed. This method implements a hybrid parallel technique dependent upon BC (HP-B) to make an allocated deep consensus learning system. Javeed et al. [19] introduced an innovative Software-Defined Networking

(SDN) orchestrated DL method to develop an intellectual IDS. The system initially deliberated the SDN framework as an appropriate decision that allows reconfiguration through static network substructure and manages the distribution model of an intelligent CE network. Afterwards, a DL-enabled IDS that applies Cuda-enabled Bi directional Long Short-Term Memory (Cu-BLSTM) was presented.

Nguyen et al. [20] developed protective intrusion recognition with BC-enabled data transmission with a classifier method in the CPS platform. This method executes data acquisition procedures utilizing sensor devices, and intrusion detection occurs using DBN architecture. Moreover, the introduced method uses a multiple share creation (MSC) technique. Further, the BC technology that performs the ResNet-based classification system was implemented. Alzubi et al. [21] developed an innovative method employing DL and BC technologies for EHR privacy-preservation. The processed database categorized abnormal and normal users exploiting the CNN method. Subsequently, with the help of BC incorporated with a cryptography-based FL method, unreliable users might be processed and extracted from the databases with the availability of the EHR. Navaneethan and Janakiraman [22] proposed a DL-based scheme named the Hybrid Interactive Autodidactic School-Based Teaching Learning Optimizer (HIASTLO) technique. The IoT-enabled e-commerce BC was utilized to extract and remove the diverse cyberattacks in the network. Implemented to increase the weight of the NNs. Additionally, MudraChain and NormaChain should be employed to evaluate this recommended task's transmission period [23], and a new architecture, such as the deep Q-learning-based NN with privacy preservation algorithm (DQ-NNPP) technique, should be developed. Yin et al. [24] present a DL-based intrusion detection model (RNN-IDS).

In [25], a model named DeepLG SecNet, employing incorporation of DL, LSTM, gated Secure Network (SecNet), and Crossover Chaos Game Optimization (CCGO) models, is proposed. Li et al. [26] introduced an edge computing-driven data collection model for smart farming, extracting key feature data types (KFDTs) from historical datasets and employing software-defined wireless sensor networks (SDWSNs) for choosing event types, safeguarding timely sensing. Padma and Ramaiah [27] present a model utilizing the trust-based ensemble consensus Grey Wolf Optimization (GWO) model. This technique also integrates PoW and PoS methods for improved trust levels. The model also implements sidechaining with the effectual GWO model. Ramaiah et al. [28] highlight Industry 4.0, aiming to enhance smart manufacturing effectually via technologies such as AI, Big Data, and IoT, but encounter safety threats needing robust solutions such as BC. In [29], the SecPrivPreserve framework is proposed, employing BC to safeguard IoT data. It implements diverse safety measures and Chebyshev polynomials for data protection. Shingari and Mago [30] presents β -CAS, employing a novel access control method (β), dual-layer neural networks, and IoT

records to prevent unauthorized access and restore control in closed IoT-consumer electronics loops effectually. Li et al. [31] presents a reputation-based stable BC-based model, incorporating dynamic sharding alterations based on a defined stability analysis and utilizing deep reinforcement learning (DRL) method. In [32], a permissions-based BC framework, integrating hybrid DL methods is proposed. Sasikumar et al. [33] proposes a secure model incorporating a consensus-based access control mechanism integrating BC, digital twin technology, and real-time device surveillance.

III. THE PROPOSED MODEL

This research presents a BCAM-DLTM technique for securing consumer electronics devices in the IoT ecosystems. The BCAM-DLTM approach mainly follows a two-phase procedure: access management and threat detection. Fig. 1 demonstrates the workflow of the BCAM-DLTM approach.

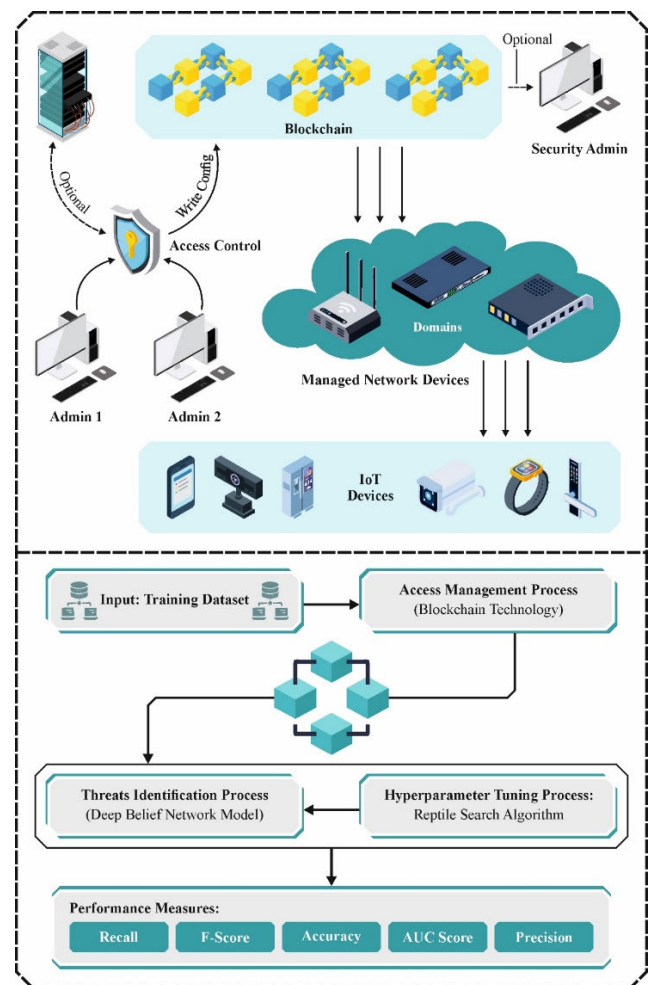


FIGURE 1. Workflow of BCAM-DLTM approach.

A. BC TECHNOLOGY

BC technology can be applied to access management of consumer electronics devices using the BCAM-DLTM technique. BC technology offers a method to generate consensus

systems without believing other nodes in a distributed system [34].

Depending upon BC, every device can manage itself without physical care. When the equipment still occurs, the lifecycle of the entire network becomes highly extensive, and the process price is considerably decreased. For instance, everyday domestic objects can naturally and mechanically perform economic actions with other substances or the external world. Smart meters could simplify extra promising current bills by modifiable energy utilization, frequency, etc. BC is very appropriate for access management and authentication. The ABE model projects uniting IoT with BC technology; safe access management in this research was employed to resolve the individuality and access tasks of connecting smart devices in IoT networks. The model was separated into three fragments: IoT, user, and BC method. The BC model comprises baseline stations that uphold a united BC but cannot be altered. The intelligent contract protected in the BC identifies the foremost access control procedure. The BC plays a vital part in the initialization procedure, which is utilized for consumer authentication and registration, BC management and control, etc. The exact execution stages of IoT with BC are defined below.

Initialization: Consumers are separated into ordinary consumers and device owners. By executing the digital signature process, a pair of public-secret key pairs, i.e. $\text{Key}=\{(\text{pk1},\text{sk1}), (\text{pk2},\text{sk2}), \dots, (\text{pkx},\text{skx})\}$ can be obtained. Then, the BC model produces public-secret key pairs for every consumer.

Device marking: Consumers include their devices in the BC method. By executing hashing processes on the owner's public key, every device is allocated an exclusive value $g = \text{hash}(\text{pk})$. In this method, every device is recognized and signified as (pk, g) . The pk denotes the public key, and the device signifies g .

Transaction generation: The owners with rights to these smart gadgets can handle and function them over system confirmation by their secret and public keys. Simultaneously, they create a transaction tx . By utilizing their secret key sk , they produce a signature e . So, this access demand will be loaded up to the distributed system as a transaction tx .

Verification: Miners acquire this transaction tx and validate the signature e by consistent owner's pk . If the signature e is accurate, it indicates that this gadget's holder has produced it. Otherwise, this transaction tx is prohibited.

Block generation: Miners contain an accessible transaction tx into a novel block. Over the consensus device, like PoS, miners contest for the right to include this novel block in the chain.

Access: Miners approve a general set of authenticated dealings in the ledger. Also, the miner who acquires the right to create a novel block will be prized. After that, the block has been decided by complete system nodes, transaction tx will be broad, and the consumer can access this gadget. His public key for confidentiality security creates

the address of the consumer's account. Public and secret keys are correspondingly employed for data encryption and confirmation. Initially, every communication among entities was required to be retained by the sender's sk and then confirmed by the receiver as per the sender's pk to ensure the message was not damaged.

B. THREAT DETECTION USING THE DBN MODEL

The BCAM-DLTM technique proficiently uses the DBN method to classify threats. As a building block of DBN, the Restricted Boltzmann machine (RBM) is primarily composed of a visible layer (VL) and a hidden layer (HL) [35]. Unlike the Boltzmann machine (BM) with a complete connection graph structure, RBM bounds the connection of peer nodes to safeguard their joint individuality. Fig. 2 illustrates the architecture of the DBN model.

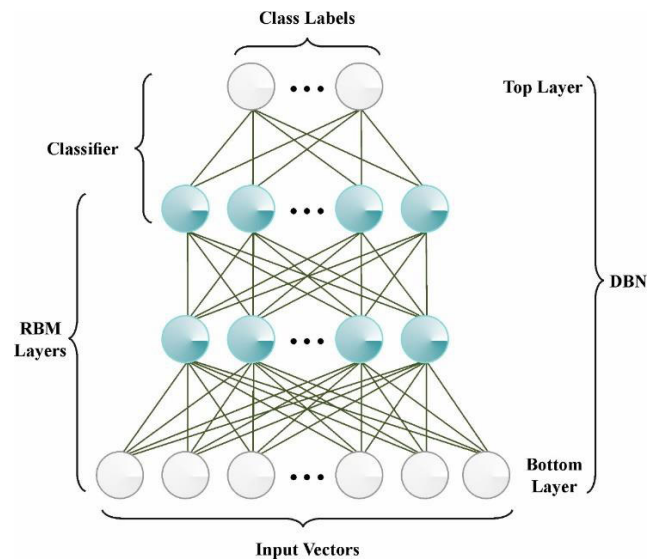


FIGURE 2. DBN architecture.

RBM is a probabilistic model where the parameter is composed of weight and bias. Assume the hidden and visible vectors are h and v . The possibility density is formulated by using Eq. (1):

$$p(v, h) = \frac{e^{-E(v, h)}}{\int \int_{v, h} e^{-E(v, h)}}, \quad (1)$$

Here, $E(v, h)$ refers to the energy function. Binary and Gaussian units are the most commonly used types. Instead of a binary-binary energy function, a better selection is to use Gaussian energy while handling continuous-value data.

$$E(v, h) = \sum_{i \in \text{vis}} \frac{(v_i - a_i)^2}{2\sigma_i^2} + \sum_{j \in \text{hid}} \frac{(h_j - b_j)^2}{2\sigma_j^2} - \sum_{i \in \text{vis}, j \in \text{hid}} \frac{v_i h_j}{\sigma_i \sigma_j} w_{ij}, \quad (2)$$

$$E(v, h) = - \sum_{i \in \text{vis}} a_i v_i - \sum_{j \in \text{hid}} b_j h_j - \sum_{i \in \text{vis}, j \in \text{hid}} v_i h_j w_{ij}, \quad (3)$$

Now h_j and v_i are the activation states of j^{th} HL and i^{th} VL units, correspondingly; b_j and a_i are the respective bias terms; w_{ij} shows the weight that links v_i and h_j . In the Gaussian energy function, σ_i and σ_j are the standard deviations of Gaussian noise presented in j^{th} HL, and i^{th} and VL units take the value as $\sigma_i = \sigma_j = 1$. The probability distribution $p(h|v), p(v|h)$ is attained using Bayesian inference with joint probability distribution $p(v, h)$ and marginal probability distribution $p(v)$ and $p(h)$,

$$p(h|v) = p(v, h) / p(v) = e^{-E(v, h)} / \int_h e^{-E(v, h)}$$

$$p(v|h) = \frac{p(v, h)}{p(h)} = \frac{e^{-E(v, h)}}{\int_v e^{-E(v, h)}} \quad (4)$$

The conditional distribution must follow the following uniform distribution for the Gaussian unit.

$$p(h_j|v) \sim N(\mu_j, \sigma_j), \mu_j = b_j + \sigma_j \sum_{i \in vis} \frac{v_i}{\sigma_i} w_{ij} = line(v)$$

$$p(v_i|h) \sim N(\mu_i, \sigma_i), \mu_i = a_i + \sigma_i \sum_{j \in hid} \frac{h_j}{\sigma_j} w_{ij} = line(h) \quad (5)$$

DBN comprises a manifold stacked RBM and a layer of output on the last RBM. The training process of DBN contains dual phases: layer-by-layer unsupervised pre-training and modification. A layer-by-layer greedy model has been used throughout the pre-training stage to train the RBM. Once finished, its HL is employed as the noticeable layer for the following RBM. This method trains RBMs individually until the previous RBM is proficient. After pre-training, a classification layer is included in the final HL, and the DBN will be modified by diminishing the mistakes in the projected output value and label. The BP technique was executed to gradually permit the fault from the final to the lowest input layer. With this method, all system parameters are upgraded.

C. RSA-BASED HYPERPARAMETER TUNING

Finally, the hyperparameter tuning procedure uses RSA to enlarge the detection results of the DBN technique. RSA is a new metaheuristic optimizer technique stimulated by the natural attack behaviour of crocodiles [36]. Crocodiles have been proficient and the most skilled hunters. The presented method is inspired mainly by the hunting and envioning procedure of the target. In RSA, a set of random candidate solutions (X) is initialized by Eq. (7) to begin the optimization algorithm. The optimal solution obtained is considered the fittest solution throughout the iteration.

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,n-1} & x_{1,n} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,n-1} & x_{2,n} \\ \vdots & \vdots & & x_{i,j} & \vdots \\ x_{N,1} & x_{N,2} & \cdots & x_{N,n-1} & x_{N,n} \end{bmatrix} \quad (6)$$

$$x_{i,j} = rand \times (UB - LB) + LB, j = 1, 2, \dots, n \quad (7)$$

In Eq. (6), $x_{i,j}$ is the j^{th} location for the solution integer $i.N$ indicates the overall amount of candidate solutions, and n refers to the size of search problems. *rand* indicates the random integer. *UB* and *LB* are the maximal and minimal limits.

1) EXPLORATION PHASE

Belly and high walking are two types of movements while surrounding the prey. Unlike another search phase (hunting time), neither movement permits the approach of the targeted prey due to the possibility of disturbance. The exploration shows a wide search range; it may discover the possible density space after particular tries. Furthermore, the exploration methods withstand the exploration phase in the search range through distributed and immeasurable research. In the RSA, the maximal iteration count is split into four equivalent choices; the 1st dual is for the exploration stage, whereas the remaining is for the exploitation stage. The exploration process of RSA search for the optimizer issue is based on the two search processes: high walk tactic, provisional by the 1st part of iterations $t \leq \frac{T}{4}$, and belly walk tactic conditioned by the 2nd quarter $\frac{T}{4} < t \leq \frac{T}{2}$. The process to update the position of the search agent is shown as:

$$x_{i,j}(t+1) = \begin{cases} best_j(t) - \eta_{i,j}(t) \times \beta - R_{i,j}(t) \times rand, & t \leq \frac{T}{4} \\ best_j(t) \times x_{r1,j} \times ES(t) \times rand, & \frac{T}{4} < t \leq \frac{T}{2} \end{cases} \quad (8)$$

In Eq. (8), $best_j(t)$ indicates the j^{th} location in the optimum solution, $\eta_{i,j}(t)$ denotes the hunting factor, β shows the sensitivity parameter for exploration accuracy control, $R_{i,j}(t)$ refers to the reduced function to reduce the search range. $r1$ indicates the arbitrary value within $[1, N]$, where N implies the number of solutions. $x_{r1,j}$ shows the random location of the j^{th} solution. $ES(t)$ represents the randomly declined probability ratio within $[2, -2]$ and is provided as:

$$x_{i,j}(t+1) = \begin{cases} best_j(t) - \eta_{i,j}(t) \times \beta - R_{i,j}(t) \times rand, & t \leq \frac{T}{4} \\ best_j(t) \times x_{r1,j} \times ES(t) \times rand, & \frac{T}{4} < t \leq \frac{T}{2} \end{cases} \quad (9)$$

$$R_{ii}(t) = \frac{best_j(t) - x_{r2,j}}{best_j(t) + \epsilon} \quad (10)$$

$$ES(t) = 2 \times r_3 \times \left(1 - \frac{1}{T}\right) \quad (11)$$

Now, ϵ indicates the insignificant value, and $r2$ represents the random value within $[1, N]$. $r3$ displays the generated value at random within $[-1, 1]$. $P_{i,j}$ refers to the ratio deviation among the positions of j^{th} candidates in the best

solution and their locations in the existing iteration.

$$P_{ij} = \alpha + \frac{x_{i,j} - M(x_i)}{best_j(t) \times (UB_j - LB_j) + \epsilon} \quad (12)$$

In Eq. (12), LB_i and UB_i are the lower and upper limits of the j^{th} location. $M(x_i)$ is the average position of the i^{th} solution. α is a sensitivity operation to control the exploration accuracy, and ϵ is attuned at 0.1.

$$M(x_i) = \frac{1}{n} \sum_{j=1}^n x_{i,j} \quad (13)$$

2) EXPLOITATION PHASE

Crocodiles have dual strategies throughout the hunting procedure: cooperation and coordination hunting. The hunting mechanisms allow crocodiles to approach the prey owing to their strengthening, unlike the encircle strategy. The hunting strategy is modelled in Eq. (14). In contrast, the hunting coordination system can be attained throughout the 3rd part of iterations $\frac{T}{2} < t \leq 3\frac{T}{4}$, and the cooperation tactic has been implemented in $3\frac{T}{4} < t \leq T$.

$$x_{i,j}(t+1) = \begin{cases} best_j(t) \times P_{i,j}(t) \times rand, \\ \frac{T}{2} < t \leq 3\frac{T}{4} \\ best_j(t) - \eta_{i,j}(t) \times \epsilon - R_{i,j}(t) \times rand, \\ 3\frac{T}{4} < t \leq T \end{cases} \quad (14)$$

Exploitation search methods (cooperation and coordination hunting) are attempting to escape from local optima. This strategy supports the exploration stage in identifying the optimum solution.

The fitness function (FF) is the significant feature that affects the performance of the RSA model. The hyperparameter range procedure includes the encoded model solution to calculate the effectiveness of the candidate solution. In this research, the RSA method reflects accuracy as the primary measure to plan the FF and is expressed as:

$$Fitness = \max(P) \quad (15)$$

$$P = \frac{TP}{TP + FP} \quad (16)$$

FP and TP signify the false and true positive values from the formula above.

IV. PERFORMANCE VALIDATION

The experimental validation of the BCAM-DLTM method is verified by employing the NSLKDD dataset [37]. Table 1 provides a detailed description of a database.

Fig. 3 proves the classifier outcomes of the BCAM-DLTM model below 70:30 of TRAP/TESP. Figs. 3a-3b represents the confusion matrices delivered by the BCAM-DLTM model. The figure signified that the BCAM-DLTM model is precisely known and classified into all five classes. Similarly, Figs. 3c-3d establishes the general classification result of

TABLE 1. Details of the dataset.

Classes	No. of Instances
Normal	77054
DoS	53385
Probe	14410
R2L	3416
U2R	252
Total Instances	148517

the BCAM-DLTM method. The outcomes demonstrate the productive performance of the BCAM-DLTM model in all cases.

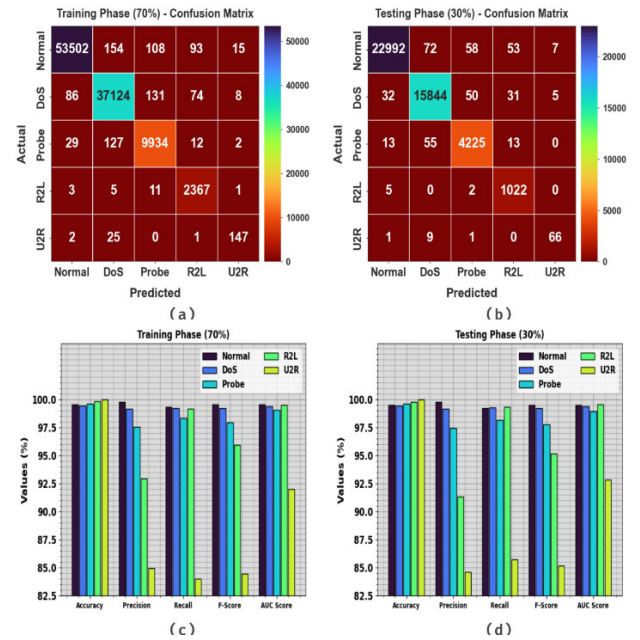


FIGURE 3. 70:30 of TRAP/TESP (a-b) Confusion matrices and (c-d) Classifier result.

The attack recognition outcomes of the BCAM-DLTM technique under 70:30 of TRAP/TESP are portrayed in Table 2 and Fig. 4. The results show the efficient performance of the BCAM-DLTM under all cases. With 70% of TRAP, the BCAM-DLTM technique offers average $accu_y$, $prec_n$, $reca_l$, F_{score} , and AUC_{score} of 99.66%, 94.88%, 96.00%, 95.42%, and 97.88%, respectively. Meanwhile, with 30% of TESP, the BCAM-DLTM model delivers average $accu_y$, $prec_n$, $reca_l$, F_{score} , and AUC_{score} of 99.63%, 94.46%, 96.32%, 95.36%, and 98.04%, respectively.

The $accu_y$ curves for training (TRA) and validation (VL) are visible in Fig. 5 for the BCAM-DLTM model, providing valuable visions into its performance under several epochs. Mainly, there is a reliable development in both TRA/TES $accu_y$ for rising epochs, showing the model's skill in detecting and learning designs from both TRA/TES data. The ascending trend in TES $accu_y$ underlines the model's

TABLE 2. Attack recognition outcome of BCAM-DLTM technique at 70:30 of TRAP/TESP.

Classes	$Accu_y$	$Prec_n$	$Recal_l$	F_{Score}	AUC_{Score}
TRAP (70%)					
Normal	99.53	99.78	99.31	99.54	99.54
DoS	99.41	99.17	99.20	99.19	99.37
Probe	99.60	97.55	98.32	97.93	99.03
R2L	99.81	92.93	99.16	95.95	99.49
U2R	99.95	84.97	84.00	84.48	91.99
Average	99.66	94.88	96.00	95.42	97.88
TESP (30%)					
Normal	99.46	99.78	99.18	99.48	99.47
DoS	99.43	99.15	99.26	99.20	99.39
Probe	99.57	97.44	98.12	97.78	98.92
R2L	99.77	91.33	99.32	95.16	99.55
U2R	99.95	84.62	85.71	85.16	92.84
Average	99.63	94.46	96.32	95.36	98.04

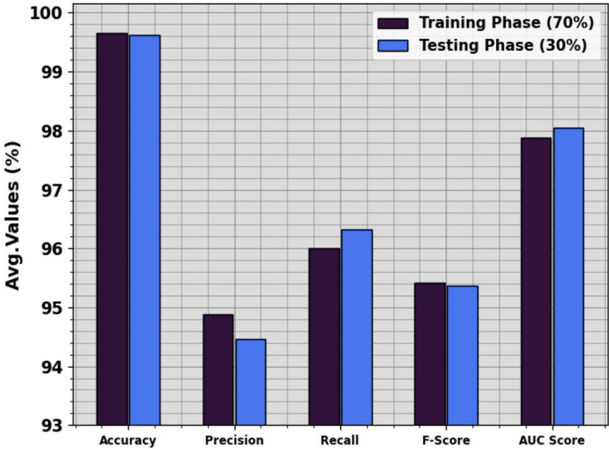


FIGURE 4. Average of BCAM-DLTM technique at 70:30 of TRAP/TESP.

flexibility relative to the TRA dataset and its capacity to make precise forecasts on unseen data, emphasizing generalized solid abilities.

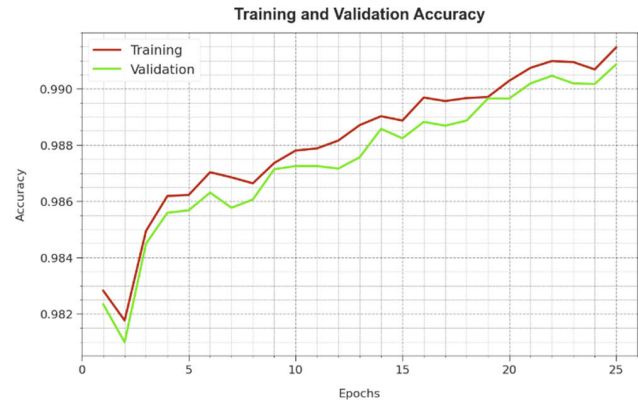


FIGURE 5. Accu_y curve of the BCAM-DLTM technique.

Fig. 6 summarizes the TRA/TES loss values for the BCAM-DLTM method across numerous epochs. The TRA loss reliably reduces as the model improves its weights to minimize classification faults on both datasets. The loss curves demonstrate the model's location with the TRA data, highlighting its capacity to take patterns proficiently in both datasets. Note the nonstop modification of parameters in the BCAM-DLTM method, which is expected to minimize differences between predictions and real TRA labels.



FIGURE 6. Loss curve of the BCAM-DLTM technique.

Regarding the precision-recall (PR) curve in Fig. 7, the results indicate that the BCAM-DLTM technique steadily attains enhanced PR values through each class. These outcomes underline the model's capability for discerning among diverse classes and its efficiency in distinguishing classes.

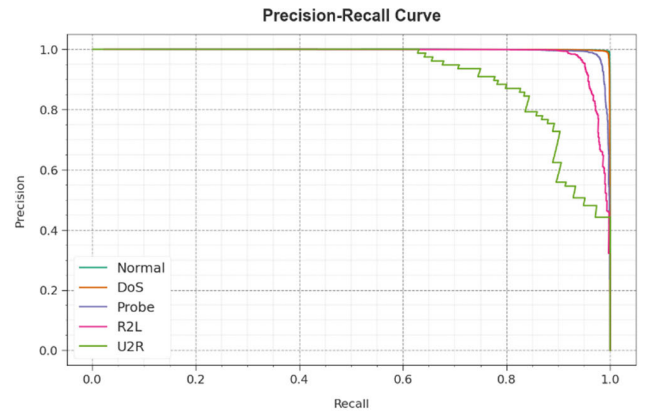


FIGURE 7. PR curve of the BCAM-DLTM technique.

Besides, in Fig. 8, ROC curves generated by the BCAM-DLTM method, signifying its ability to differentiate amongst classes, are presented. These curves deliver valuable visions into how the trade-off between TPR and FPR differs through diverse classification epochs and thresholds. The consequences highlight the model's accurate identification

performance below several classes, underlining its efficacy in addressing assorted classification tasks.

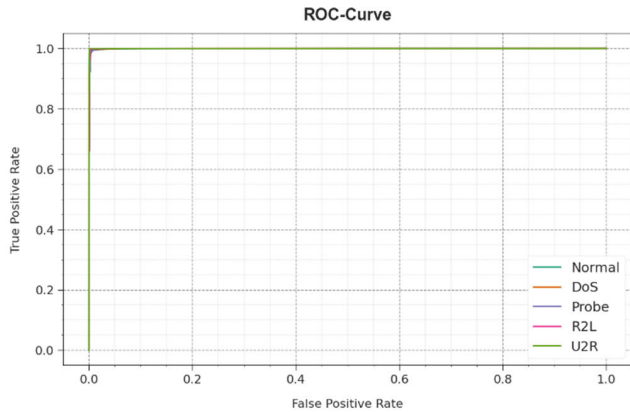


FIGURE 8. ROC curve of the BCAM-DLTM technique.

Table 3 provides a brief comparison study of the BCAM-DLTM model [38], [39], [40]. Fig. 9 inspects a comparative analysis of the BCAM-DLTM technique regarding $accu_y$. Based on $accu_y$, the BCAM-DLTM technique gains a higher $accu_y$ of 99.63% while the NIDSML, AEML-ID, NSLKDD-IoT, DL-Improved ID, DL-NIDS, AD-NTELM, DLDCSCA-IoTCN, RKOA-AEID, HSCCOFS-DL, ABC-AFS, GSGW-DT-DT, and HBAOH-DBN approaches obtain lower $accu_y$ of 83.84%, 85.33%, 78.64%, 94.74%, 88.54%, 91.83%, 98.34%, 98.94%, 99.19%, 98.60%, 99.02%, and 99.25%, respectively.

TABLE 3. Comparative analysis of BCAM-DLTM method with recent models [38], [39], [40].

Research	$Accu_y$	$Prec_n$	$Reca_l$	F_{Score}
NIDSML	83.84	71.13	70.04	70.38
AEML-ID	85.33	73.60	72.49	71.54
NSLKDD-IoT	78.64	72.07	71.02	71.18
DL-Improved ID	94.74	71.05	69.32	74.43
DL-NIDS	88.54	71.16	70.43	70.44
AD-NTELM	91.83	73.44	70.98	74.09
DLDCSCA-IoTCN	98.34	72.15	72.92	73.34
RKOAAEID	98.94	75.33	86.45	79.52
HSCCOFS-DL	99.19	85.20	87.72	86.87
ABC-AFS	98.60	86.10	76.80	77.57
GSGW-DT-DT	99.02	84.19	78.61	80.04
HBAOH-DBN	99.25	76.41	75.19	75.78
BCAM-DLTM	99.63	94.46	96.32	95.36

Fig. 10 examines a comparative study of the BCAM-DLTM method regarding $prec_n$, $reca_l$, and F_{score} . Based on $prec_n$, the BCAM-DLTM method gains developed $prec_n$ of 94.46% while the NIDSML, AEML-ID, NSLKDD-IoT, DL-Improved ID, DL-NIDS, AD-NTELM, DLDCSCA-IoTCN, RKOA-AEID, HSCCOFS-DL, ABC-AFS, GSGW-DT-DT, and HBAOH-DBN techniques get lower $prec_n$ of 71.13%,

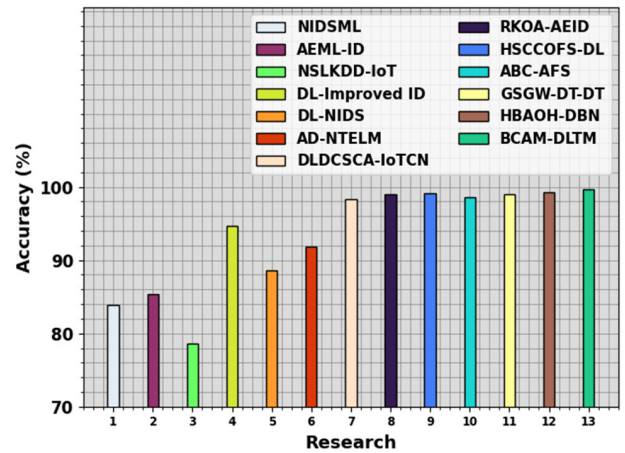


FIGURE 9. $Accu_y$ analysis of BCAM-DLTM model with recent methods.

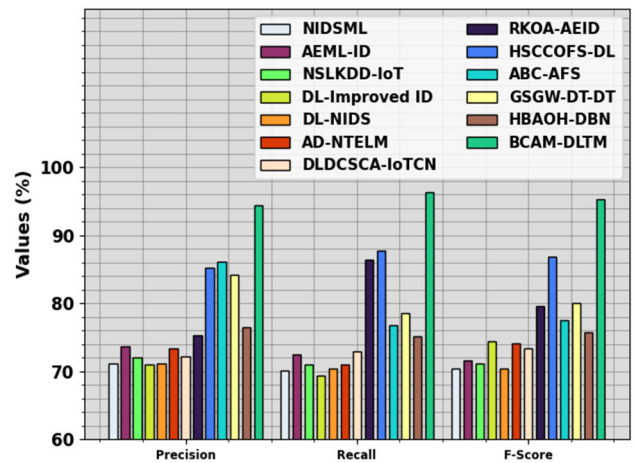


FIGURE 10. Comparative analysis of BCAM-DLTM technique with recent methods.

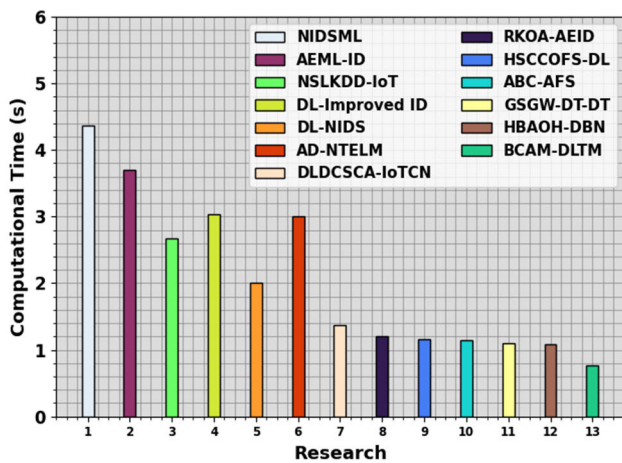
73.60%, 72.07%, 71.05%, 71.16%, 73.44%, 72.15%, 75.33%, 85.20%, 86.10%, 84.19%, and 76.41%, individually. Moreover, based on $reca_l$, the BCAM-DLTM method increased higher $reca_l$ of 96.32%. In contrast, the NIDSML, AEML-ID, NSLKDD-IoT, DL-Improved ID, DL-NIDS, AD-NTELM, DLDCSCA-IoTCN, RKOA-AEID, HSCCOFS-DL, ABC-AFS, GSGW-DT-DT, and HBAOH-DBN methods attain lower $reca_l$ of 70.04%, 72.49%, 71.02%, 69.32%, 70.43%, 70.98%, 72.92%, 86.45%, 87.72%, 76.80%, 78.61%, and 75.19%, correspondingly. Finally, based on the F_{score} , the BCAM-DLTM approach gains a higher F_{score} of 95.36%, whereas the NIDSML, AEML-ID, NSLKDD-IoT, DL-Improved ID, DL-NIDS, AD-NTELM, DLDCSCA-IoTCN, RKOA-AEID, HSCCOFS-DL, ABC-AFS, GSGW-DT-DT, and HBAOH-DBN models acquire lower F_{score} of 70.38%, 71.54%, 71.18%, 74.43%, 70.44%, 74.09%, 73.34%, 79.52%, 86.87%, 77.57%, 80.04%, and 75.78%, respectively.

Table 4 and Fig. 11 briefly compare the computational time (CT) study of the BCAM-DLTM approach. The

TABLE 4. CT analysis of BCAM-DLTM technique with recent methods.

Research	CT (s)
NIDSML	4.36
AEML-ID	3.70
NSLKDD-IoT	2.68
DL-Improved ID	3.03
DL-NIDS	2.00
AD-NTELM	3.01
DLDCSCA-IoTCN	1.37
RKOA-AEID	1.20
HSCCOFS-DL	1.16
ABC-AFS	1.14
GSGW-DT-DT	1.10
HBAOH-DBN	1.08
BCAM-DLTM	0.76

experimental value stated that the BCAM-DLTM technique beat optimal performances equated with other models. Based on CT, the BCAM-DLTM method gains smaller CT of 0.76s whereas the NIDSML, AEML-ID, NSLKDD-IoT, DL-Improved ID, DL-NIDS, AD-NTELM, DLDCSCA-IoTCN, RKOA-AEID, HSCCOFS-DL, ABC-AFS, GSGW-DT-DT, and HBAOH-DBN methods acquire more excellent CT of 4.36s, 3.70s, 2.68s, 3.03s, 2.0s, 3.01s, 1.37s, 1.20s, 1.16s, 1.14s, 1.10s, and 1.08s correspondingly.


FIGURE 11. CT analysis of BCAM-DLTM technique with recent models.

These results showcased the superior performance of the BCAM-DLTM method.

V. CONCLUSION

This study presents a BCAM-DLTM technique for securing consumer electronics devices in the IoT ecosystems. The BCAM-DLTM technique mainly monitors a two-phase procedure: access management and threat detection. In the BCAM-DLTM technique, BC technology can be applied to manage access to consumer electronics devices. Besides, the

BCAM-DLTM technique uses the DBN method proficiently to classify threats. To enhance the recognition outcomes of the DBN technique, the hyperparameter tuning process takes place using RSA. The experimental outcome study of the BCAM-DLTM approach takes place employing a benchmark dataset. The comprehensive results identified that the BCAM-DLTM approach attains superior effects over existing methodologies regarding distinct metrics. The BCAM-DLTM approach encounters threats in scalability, resource usage, and privacy. Future studies may focus on optimizing resources, improving privacy, fortifying against adversarial attacks, and performing real-world deployments for validation.

ACKNOWLEDGMENT

The authors extend their appreciation to the Deanship of Research and Graduate Studies at King Khalid University for funding this work through Large Research Project under grant number RGP2/43/45. Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2024R411), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia. The authors extend their appreciation to the Deanship of Scientific Research at Northern Border University, Arar, KSA for funding this research work through the project number NBU-FFR-2024-3030-03. This study is partially funded by the Future University in Egypt (FUE).

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